



Magikarp Ground Mission

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Easy

General Skills

picoCTF 2021

AUTHOR: SYREAL

Description

Do you know how to move between directories and read files in the shell? Start the container, `ssh` to it, and then `ls` once connected to begin.

Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.

Its current status is:

NOT_RUNNING

Launch Instance

Hints 

1

93,353 users solved



94%

Liked



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Author: The Analyst: Hyposelenia

Challenge: Magikarp Ground Mission

Category: General Skills

Date: 01/11/26



I. Objective

The objective of this task was to identify the hidden flag by navigating through different directories in a Linux system and reading specific text files using basic shell commands. This challenge focuses on understanding directory movement, file reading, and following instructions provided within the system itself.

II. Background

In Linux-based systems, files are organized inside directories, which are similar to folders. Unlike graphical systems where users click through folders, Linux often requires users to navigate using text-based commands in a shell. This challenge was designed to teach beginners how to move between directories, understand the root and home directories, and read files using simple commands such as **ls**, **cd**, and **cat**.

III. Tool Used

- **Oracle VirtualBox (Kali Linux)** – Used as the Linux environment to run the challenge
- **Linux Terminal** – Used to type commands and interact with the system
- **ssh** – Used to securely connect to a remote computer. Like opening a door to another computer far away and controlling it using text
- **cd** – Used to change directories (move between folders)
- **ls** – Used to list files and folders
- **cat** – Used to read the contents of text files
- **-p:** Stands for “port,” which is like choosing the correct door number to enter
- **ctf-player@hostname:** Means “log in as user ctf-player on a computer named hostname”

IV. Methodology

1. Kali Linux was launched using Oracle VirtualBox.
2. The following SSH command was used to connect to the remote challenge server:

```
ssh ctf-player@wily-courier.picoctf.net -p 64233
```



3. After accepting the fingerprint by typing yes, the password **8c606eb1** was entered. (**Note: the password does not appear while typing, which is normal for security reasons.**)
4. Once logged in, the `ls` command was used to list files in the current directory. Two files appeared:

1of3.flag.txt

instruction-to-2-3.txt

5. The `cat` command was used to read both files.
6. **1of3.flag.txt** revealed the first part of the flag:

picoCTF{xxsh_

7. The instruction file stated: “**Next, go to root of all things, more succinctly ‘/’**”.
8. The command “**cd /**” was used to move to the root directory.
9. Using `ls` in the root directory displayed several folders and files, including **2of3.flag.txt** and **instruction-to-3-3.txt**.
10. at **2of3.flag.txt** revealed the second part of the flag:

0ut_0f_//4t3r_

11. The next instruction said: “**Lastly, go home... more succinctly ‘~’**”.
12. The command **cd ~** was used to return to the home directory.
13. Inside the home directory, **cd ctf-player** was used to enter the user folder. I knew it was ctf-player since I used `ls`.
14. Using `ls`, the file **3of3.flag.txt** was found and opened using `cat`, revealing the final part of the flag:

0b24fc4f}

15. All three parts were combined to form the complete flag.



V. Result

picoCTF{xxsh_out_of_//4t3r_0b24fc4f}

```
(kali@kali)-[~]
$ ssh ctf-player@wily-courier.picoctf.net -p 64233
The authenticity of host '[wily-courier.picoctf.net]:64233 ([18.189.99.27]:64233)' can't be established.
ED25519 key fingerprint is SHA256:ErlUUvYlrAxFSW1tIdzfOnGTBSr50FkZvz0nMN4Vodw.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
  ~/.ssh/known_hosts:3: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[wily-courier.picoctf.net]:64233' (ED25519) to the list of known hosts.
ctf-player@wily-courier.picoctf.net's password:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 6.14.0-1012-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

```
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ctf-player@pico-chall$ ls
1of3.flag.txt  instructions-to-2of3.txt
ctf-player@pico-chall$ cat 1of3.flag.txt
picoCTF{xxsh_
ctf-player@pico-chall$ cat instructions-to-2of3.txt
Next, go to the root of all things, more succinctly '/'
ctf-player@pico-chall$ cd /
ctf-player@pico-chall$ ls
2of3.flag.txt  bin  boot  challenge  dev  etc  home  instructions-to-3of3.txt  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
ctf-player@pico-chall$ cat 2of3.flag.txt
0ut_0f_//4t3r_
ctf-player@pico-chall$ cat instructions-to-3of3.txt
Lastly, ctf-player, go home... more succinctly '~'
ctf-player@pico-chall$ ~
-bash: /home/ctf-player: Is a directory
ctf-player@pico-chall$ ls
2of3.flag.txt  bin  boot  challenge  dev  etc  home  instructions-to-3of3.txt  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
ctf-player@pico-chall$ cd home
ctf-player@pico-chall$ ls
ctf-player
ctf-player@pico-chall$ cd ctf-player
ctf-player@pico-chall$ ls
3of3.flag.txt  drop-in
ctf-player@pico-chall$ cat 3of3.flag.txt
0b24fc4f}ctf-player@pico-chall$ Connection to wily-courier.picoctf.net closed by remote host.
Connection to wily-courier.picoctf.net closed.
```



VI. Explanation

This challenge works like a treasure hunt inside a computer. Each folder is like a room, and each file is like a note giving instructions. The computer starts you in one room, then tells you to go to the “root,” which is the biggest room that holds everything. After that, it tells you to go “home,” which is your personal room. By following instructions and using simple commands to move and read files, all parts of the flag were found and combined. Use `cd` to move directory, use `ls` to list the present in your current directory, use `cat` to display and read contents inside the file.

VII. Conclusion

This challenge teaches the importance of understanding Linux directory structure, basic shell navigation, and reading files using terminal commands. These skills are essential for cybersecurity, system administration, and ethical hacking, as many real-world systems do not use graphical interfaces. Learning how to move efficiently through directories and follow command-line instructions builds a strong foundation for more advanced security tasks.

— **The Analyst: Hyposelenia**